Why a Metric Is Needed to Define a Gradient

What Smooth Structure Gives You

On a smooth manifold M, a smooth function

$$f : M \to \mathbb{R}$$

has a differential at each point:

$$df_p \in T_p^*M.$$

This is a covector, meaning a linear map

$$df_p : T_pM \to \mathbb{R}.$$

It eats a tangent vector X_p and returns the directional derivative

$$df_p(X_p) = X_p[f].$$

This construction uses only the smooth structure. No metric and no connection are required.

Why This Is Not Yet a Direction to Follow

A gradient flow needs a tangent vector field:

$$\dot{p}_t = -(\text{some tangent vector at } p_t).$$

But df_p is not a tangent vector. It lives in the dual space T_p^*M, not in T_pM.

The distinction matters. A covector measures directions; a vector is a direction. The differential df_p tells us how f changes along every possible direction, but it is not itself one of those directions.

Can a Connection Help?

A connection lets us differentiate vector fields along vector fields:

$$\nabla_X Y.$$

It tells us how to compare tangent vectors at nearby points.

But it does not, by itself, give a canonical map

$$T_p^*M \to T_pM.$$

So a connection alone does not turn df_p into a vector field and does not define gradient descent.

What a Riemannian Metric Adds

A Riemannian metric gives an inner product

$$g_p : T_pM \times T_pM \to \mathbb{R}.$$

For each vector $Y_p \in T_pM$, the metric produces a covector

$$g_p(Y_p, \cdot) \in T_p^*M.$$

Because g_p is nondegenerate, every covector arises uniquely in this way. Thus the metric gives an isomorphism

$$T_pM \cong T_p^*M.$$

The gradient is the vector corresponding to df_p under this isomorphism:

$$df_p(\cdot) = g_p(\operatorname{grad}_g f(p), \cdot).$$

Could One Choose Another Identification?

Yes, but it would be extra structure. A bare smooth manifold has no canonical way to identify tangent and cotangent spaces. Choosing a Riemannian metric is the standard geometric way to make that identification.

Different metrics produce different gradients and therefore different gradient flows. This is not a bug; it is the point. Gradient descent is metric-dependent.